

# Positive–Unlabeled Learning for Marine Object Detection

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**Abstract.** Underwater object detection is an important field of application in underwater exploration, marine biodiversity monitoring and environmental analysis. But often the data collected under water is not fully annotated as many marine organisms are found in an image without being labeled. The traditional object detection models assume these regions without labels as background, thus resulting in incorrect learning and poor detection accuracy. To solve this problem, this research presents a PU learning based marine object detection framework which can effectively utilizes both the labeled and unlabeled data. The system proposed consists of preprocessing, noise filtering, handling of class imbalance, pseudo-label generation and a teacher student semi-supervised learning framework in conjunction with the YOLO object detection. While the teacher model is used to pseudo-label for unlabeled images, the student model is used to learn from both the pseudo-labeled and the original images to enhance the detection performance. The purpose of the proposed approach is to boost detection recall, enhance the mean Average Precision (mAP) and correctly recognize marine organisms even if the annotations are incomplete. The proposed framework improved marine object detection performance by effectively utilizing unlabeled underwater images through pseudo-label learning. Experimental results achieved an mAP50 improvement from 0.4279 to 0.5943 and recall improvement from 0.4565 to 0.7047 compared to supervised learning. The proposed framework is evaluated on the underwater FathomNet dataset, and the results are presented in a tabular and graphical format to show how the proposed framework outperforms the traditional supervised learning algorithms.

**Keywords:** :Marine Object Detection, Positive–Unlabeled Learning, Semi-Supervised Learning, Pseudo-Labeling, YOLO, Underwater Image Analysis.

## I Introduction

A field of research of paramount importance in underwater exploration, marine ecosystem monitoring, biodiversity analysis and oceanography is marine object

detection. Underwater image processing is used for identifying marine life to allow scientists to study aquatic ecosystems, track endangered species and gain insights into environmental shifts. Underwater imaging technologies are rapidly advancing and vast amounts of marine image data is becoming available to be automatically analysed with deep learning techniques.

Deep learning object detection approaches like Faster R-CNN, YOLO, and DETR have shown good results in computer vision tasks, however, the detection of the objects under water is still difficult. In marine datasets, the images are frequently blurred, the lighting conditions are low, there is water distortion, occlusion, and a class imbalance issue. Furthermore, underwater data is often only partially annotated, in which only a subset of organisms is annotated by experts while other visible objects are not.

Traditional supervised learning methods make the assumption that all unlabeled areas are part of the background class. This assumption generates false learning signals, as unlabeled marine organisms are assumed to be negative samples during training. Consequently, the model cannot detect some critical species in the marine environment and the accuracy rate of the model is reduced. This problem can be considered a Positive-Unlabeled (PU) learning problem, where only positive labeled samples and unlabeled samples are available.

To address these problems, this study presents a Positive-Unlabeled learning framework for marine object detection. This proposed system consists of a combination of pre-processing, noise filtering, handling of class imbalance, pseudo label generation and a teacher-student semi-supervised learning framework. The teacher model is first trained with the labeled underwater images. The teacher model is then used to predict high-confidence detections on unlabeled images and pseudo-labels are created. These pseudo-labeled samples are then mixed up with the original labeled data set and used to train the student model effectively to learn from the incomplete annotations.

The proposed framework is based on the YOLO-based object detection because of its high detection speed and strong localization performance. The primary goal of this research is to design a robust marine object detection model that can detect accurately the underwater organisms from both labeled and unlabeled data and can enhance the mean Average Precision (mAP), recall and detection efficiency.

## II Literature Survey

Guo et al. suggested a Positive-Unlabeled (PU) learning-based data purification framework for object detection with the combination of small labeled and large-scale unlabeled datasets. The method employs a PU classifier to select useful samples from a huge unlabeled dataset like ImageNet and then the data set is merged with labeled COCO data to training. The framework also proposes self-distillation approaches such as hint learning and ground-truth bounded knowledge distillation to boost feature learning and detection performance. The experimental results indicated that the proposed approach achieved better per-

formance than the baseline Feature Pyramid Network (FPN) model, with a performance gap of 39.4% mAP to 37.4% mAP on the COCO dataset. However, the method necessitates having a large scale unlabeled dataset and multiple training stages which raises the computational complexity and the length of training [1].

Yang et al. did not use a traditional Positive-Negative classification problem, but rather considered the object detection problem as a Positive-Unlabeled learning problem. The authors took the Region Proposal Network (RPN) of Faster R-CNN and replaced its cross-entropy loss with a PU loss function, enabling the regions that are not labeled as background or unknown. To combat overfitting and boost its robustness in cases where the annotations are limited, a non-negative PU risk estimator was also introduced. It was tested on datasets like PASCAL VOC, MS COCO, Visual Genome, and DeepLesion, and it outperforms traditional supervised approaches with respect to the mean Average Precision (mAP). This approach, however, poses a computational burden and cannot solve the problem of class imbalance and real-time detection completely [2].

Xu et al. introduced a semi-supervised object detection approach by pseudo labeling and teacher-student learning. It is combined with unlabeled images, where the teacher model produces pseudo-labels for unlabeled images, and the student model is trained from both of them. Several data augmentation methods like scaling, translation, brightness change and rotation were also used to enhance the model’s generalization. Experimental results on the MS COCO dataset demonstrated that the proposed method boosted the performance of Faster R-CNN from 40.9 mAP to 44.5 mAP. However, the framework relies heavily on the quality of the pseudo-labels, and wrong pseudo-labels can amplify inaccuracies and impede accuracy [3].

Tan and Zheng introduced a semi-supervised learning (SSL) with Active learning (AL) approach for object detection with the base detector as Faster R-CNN. The method is based on the use of active learning techniques (Least Confidence (LC), Margin Sampling (MS), Entropy Sampling (ES)) to select informative unlabeled samples that are subsequently combined with the labeled samples to retrain. The framework was validated on several benchmarks – PASCAL VOC, MS COCO, PlantDoc, PKLot and Oxford Pets – and demonstrated object detection accuracy improvements. The method has superior performance on the mAP for various datasets, while repeated training rounds and annotations by human labour involve high computational cost and labour [4].

Sharma suggested the semi-supervised and self-supervised learning approach with YOLOv5, SimCLR, and PAWS learning techniques for underwater object detection. First, it conducts self-supervised pretraining on underwater images with no labels, in order to build visual features, and later fine-tunes the detector with labeled marine images. It was evaluated on the ROV and i2MAP underwater datasets and obtained better performance than standard supervised methods with around 78.3% mAP on the ROV dataset. It is a method that improves underwater detection accuracy, but it is expensive because the method needs a

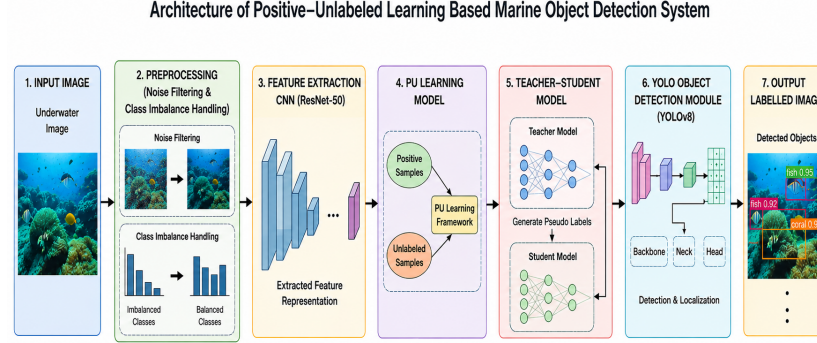
large number of unlabeled data and rich computing resources for pre-training [5].

In incremental object detection tasks, where some grounding is missing, Cermelli et al. suggested an approach named Modeling Missing Annotations (MMA). The framework employs a teacher–student knowledge distillation method to suppress labeling of unlabeled objects as background as well as Faster R-CNN. The method allows the model to retain its previously learned classes while continually adding new learned classes. The results on PASCAL VOC 2007 and Pascal SBD 2012 demonstrated an improvement of 2–6% mAP compared to the other incremental learning methods like ILOD and Faster ILOD. The framework is, however, largely concerned with two-stage detectors, and adds on extra computation time as a result of the teacher–student training approach [6].

### III Methodology

Then a proposed methodology is discussed, which is a Positive–Unlabeled (PU) learning based framework for marine object detection in underwater datasets with both labeled and unlabeled marine objects as shown in Fig. 1. The framework integrates supervised learning and pseudo-label generation, teacher–student semi-supervised learning, and YOLO object detection method to enhance the accuracy of marine species detection. Firstly, the model is trained with labeled underwater photos, and then the high confidence prediction made by the model on unlabeled underwater photos is transformed into a pseudo-label and placed in the training set. The proposed system is able to mitigate effects of incomplete annotations and enhance the detection performance in the complex underwater environment.

The methodology starts with obtaining underwater images from FathomNet dataset[7] of marine organisms with both labeled and unlabeled images. In the first phase, the model for object detection is trained with only labeled images from underwater scenes and then preprocessed to enhance the generalization of the model via image resizing, normalization, data cleaning, and augmentation techniques such as rotation, scaling, brightness adjustment and flipping. After the supervised learning, the trained model is used to predict the visual features (shapes, textures, colors, and structural patterns) of marine organisms including fish, crabs, jellyfish, sea stars and anemones from underwater images without labels. The detector infers object classes, bounding boxes and confidence levels for objects it doesn’t recognize in the ocean environment. In the second step, original labeled data and pseudo labeled data are used to retrain the model by using the teacher–student semi-supervised learning framework, while the low confidence predictions are discarded to reduce their noise. The teacher model generates reliable pseudo-labels and the student model learns from labeled data and pseudo-labels data of underwater images. To prevent the misclassification of unlabeled marine organisms as background, Positive–Unlabeled learning is incorporated. Finally, the trained model is assessed using performance metrics: mAP50, mAP50–95, precisions, recall, and F1-score. The results of the exper-

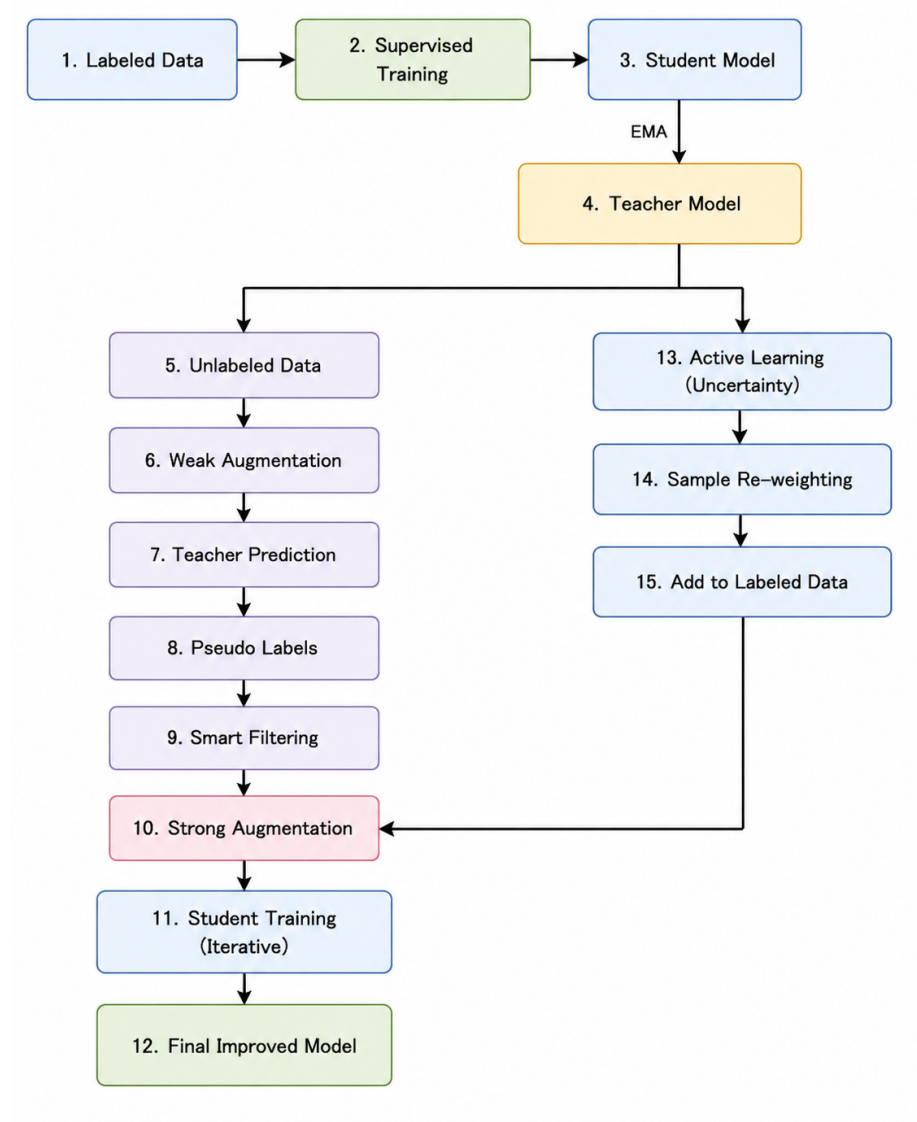


**Fig. 1.** Architecture of the proposed positive-unlabeled learning framework for marine object detection showing preprocessing, feature extraction using ResNet-50, PU learning, teacher-student semi-supervised learning, YOLO-based object detection, and final labeled underwater image generation.

iments prove that the proposed methodology has a higher accuracy of marine object detection and higher recall rates than existing supervised learning methods.

**PU Learning:** When the set of data contains only positive labeled samples and many unlabeled samples, Positive-Unlabeled (PU) learning is a machine learning technique used. In the marine object detection, some marine objects are annotated, and many of the visible marine objects are not annotated. These unlabeled objects have traditionally been considered as background noise in supervised learning and are responsible for lower detection accuracy. PU learning solves this problem by considering unlabeled samples as “unknown” instead of negative samples. Both labeled and unlabeled marine objects are used for the model’s training and pseudo labels are generated for high confidence predictions. This facilitates recall, object detection accuracy and utilization of incomplete underwater datasets.

**Teacher-Student Model:** It is a semi-supervised learning approach that enhances the object detection based on unlabeled data as shown in Fig.2. In this method, the teacher model is initially trained with the labelled underwater images. The trained teacher model then predicts object classes, bounding boxes and confidence scores for unlabeled images. Pseudo-labels are those predictions with high confidence and are included in the training set. Then pseudo-labeled data are mixed with the original labeled data for training the student model. The student model can be strengthened by repeated learning and enhance marine object detection performance. In this way, the model’s learning can be extended to more features of the objects in the underwater environment while avoiding the need for manual annotations of all the images.



**Fig. 2.** Teacher-Student semi-supervised learning architecture for marine object detection using labeled and unlabeled underwater images.

**Stage 1: Supervised Learning** In Stage 1, underwater labeled images are used to train the object detection model. This model is trained to detect various features of marine objects like shape, texture, color, and object boundaries from the annotated data. When training the detector, it can recognize marine life and anticipate their object categories and bounding boxes. However, there are many underwater datasets that have incomplete annotations in which some of the visible marine organisms are not annotated. However the traditional supervised learning techniques ignore these objects without any label and impose restrictions on the learning ability of the model and its performance in detection.

**Stage 2: Pseudo-Label Learning** In Stage 2, the trained model in Stage 1 is employed as a teacher model to predict marine objects in unlabeled underwater images. The model gives pseudo label to objects with high score. Thereafter, these pseudo-labeled images are merged with the original labeled data and fed to the student model for training. Positive-Unlabeled learning makes the model learn how to label unlabeled objects as unknown samples rather than background. The stage helps the framework to learn from both labeled and unlabeled underwater data, which enhances the model’s recognition of more marine organisms and complex underwater features.

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**Algorithm 1** Positive-Unlabeled Learning-Based Marine Object Detection

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**Require:** Labeled underwater dataset  $D_L$ , unlabeled underwater dataset  $D_U$

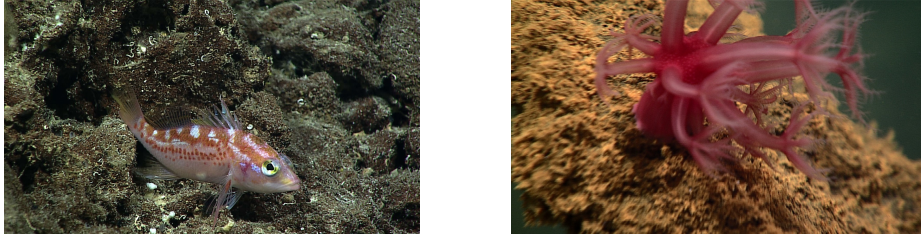
**Ensure:** Trained marine object detection model with improved detection accuracy

- 1: Preprocess underwater images using resizing, normalization, and augmentation
  - 2: Train YOLO object detector using labeled dataset  $D_L$
  - 3: Use trained teacher model to predict marine objects on unlabeled dataset  $D_U$
  - 4: **for** each unlabeled underwater image **do**
  - 5:     Predict object classes, bounding boxes, and confidence scores
  - 6:     **if** confidence score > threshold **then**
  - 7:         Generate pseudo-labels
  - 8:     **end if**
  - 9: **end for**
  - 10: Combine pseudo-labeled data with original labeled dataset
  - 11: Train student model using labeled and pseudo-labeled underwater images
  - 12: Apply Positive-Unlabeled learning to treat unlabeled objects as unknown samples
  - 13: Evaluate model performance using mAP50, mAP50-95, precision, recall, and F1-score
  - 14: **return** Final trained marine object detection model
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## A Dataset Description

The proposed marine object detection system is based on FathomNet underwater dataset[7] consisting of a vast dataset of annotated marine images collected from the underwater environment. The dataset consists of labeled and unlabeled underwater images of different marine organisms like fish, crab, jellyfish, sea star, sea cucumber, anemone, octopus, and other deep-sea organisms. The training set

is made up of about 6,463 images of the ocean with partial annotations that have box coordinates and labels only for some of the marine organisms, but not for many of the other visible objects. The test set includes some 1,425 completely annotated images for evaluation purposes. The dataset has been reduced to 32 classes of marine objects for morphological and taxonomic similarity reasons, and it is also incomplete and includes some unlabeled positive samples; thus it is very appropriate for research on Positive–Unlabeled learning and semi-supervised object detection. Additionally, the underwater images are captured under low-light, visually similar marine organisms, occlusion, visually similar background, class imbalance, and blurry background conditions, posing challenges to low-light detection, visually similar object detection, occlusion detection, class imbalance, and visually similar background detection, making the dataset a good choice to test robust underwater marine object detection frameworks [7].



**Fig. 3.** Sample underwater images from the FathomNet dataset showing marine organisms under challenging underwater environments.

## B Features of the FathomNet Dataset

The FathomNet dataset is an extensive set of underwater images for the purposes of marine object detection and biodiversity analysis. It has labeled and unlabeled images of the underwater scenes along with the corresponding labels and bounding-boxes in COCO format. This data contains approximately 6,463 training images and 1,425 test images of 32 different object classes in the marine environment, including fish, crab, jellyfish, sea star, and octopus and sample images are shown in Fig.3. It also includes difficult underwater situations like low-light conditions, blur, occlusion, and class imbalance making it appropriate for research on Positive–Unlabeled learning and semi-supervised object detection.

**Table 1.** Stage 1 Supervised Learning Evaluation Results

| Metric    | Value  |
|-----------|--------|
| mAP50     | 0.4279 |
| mAP50-95  | 0.3246 |
| Precision | 0.6283 |
| Recall    | 0.4565 |
| F1-Score  | 0.5288 |

**Table 2.** Top-5 Classes by AP50 in Stage 1

| Class      | AP50   |
|------------|--------|
| pyrosome   | 0.9950 |
| larvacean  | 0.8150 |
| urchin     | 0.8133 |
| crab       | 0.7877 |
| soft coral | 0.7245 |

## IV Results and Discussions

### Quantitative Results

#### A Stage 1 Supervised Evaluation Results

**Table 3.** Bottom-5 Classes by AP50 in Stage 1

| Class        | AP50   |
|--------------|--------|
| benthic worm | 0.0000 |
| sea pen      | 0.0000 |
| sea slug     | 0.0000 |
| sea snail    | 0.0000 |
| sea squirt   | 0.0000 |

The first supervised training stage results shown in Table.1 and Table.2 and Table.3 and achieved an mAP50 score of 0.4279 and a mAP50 – 95 score of 0.3246, showing moderate performance in object detection on the marine dataset. The model achieved a precision of 0.6283, recall of 0.4565, and an F1-score of 0.5288, indicating that the detector was able to identify several marine organisms but still failed to identify many unlabeled or difficult ones. Pyrosome had the highest AP50 score of 0.9950, followed by larvacean (0.8150), urchin (0.8133), crab (0.7877), and soft coral (0.7245) which showed good performance on classes that are highly visual. For some classes, however, including benthic worm, sea pen, sea slug, sea snail, and sea squirt, the score was  $AP50 = 0.0000$ , suggesting that the results were poor because there were few training images or class

imbalance or incompatible annotations. The results underscore the challenges of fully supervised learning on partially labeled underwater datasets, and inspire the use of Positive-Unlabeled learning and pseudo-labeling methods to enhance the accuracy of marine object detection.

## B Stage 2 Pseudo-Label Evaluation Results

**Table 4.** Stage 2 Pseudo-Label Learning Evaluation Results

| Metric    | Value  |
|-----------|--------|
| mAP50     | 0.5943 |
| mAP50-95  | 0.4366 |
| Precision | 0.5526 |
| Recall    | 0.7047 |
| F1-Score  | 0.6194 |

**Table 5.** Top-5 Classes by AP50 in Stage 2

| Class    | AP50   |
|----------|--------|
| barnacle | 0.9950 |
| pyrosome | 0.9950 |
| sea pen  | 0.9950 |
| sea slug | 0.9950 |
| octopus  | 0.8214 |

**Table 6.** Bottom-5 Classes by AP50 in Stage 2

| Class        | AP50   |
|--------------|--------|
| bivalve      | 0.2900 |
| benthic worm | 0.2450 |
| isopod       | 0.1650 |
| amphipod     | 0.0000 |
| sea squirt   | 0.0000 |

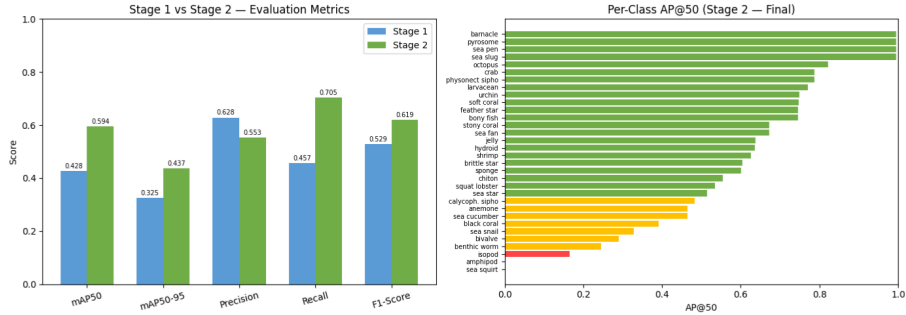
The Stage 2 pseudo-label learning evaluation results shown in Table.4 and Table.5 and Table.6 significantly outperformed the first stage (based on supervised training). The model achieved mAP50 score 0.5943 and mAP50-95 score 0.4366, indicating that the model’s overall object detection performance improved by

leveraging pseudo-labeled data. The recall saw a significant rise to 0.7047, a result that meant the model was able to identify more marine organisms—even those that had not been labeled before. Sea slugs, octopuses, sea pen and barnacles have attained very high AP50 scores of 0.9950, demonstrating strong detection ability of several marine species after pseudo-label learning, while sea slug, barnacles, pyrosomes, and sea pen have very high AP50 scores, and sea pen’s score is 0.9838. Classes like amphipod or sea squirt, however, yielded extremely poor AP50 scores, and the results were reasonably poor for bivalve, benthic worm, or isopod due to limited training samples and class imbalance problems. In general, the outcomes show that pseudo-labeling and Positive-Unlabeled learning techniques yield significant improvements in marine object detection accuracy and recall over the current supervised learning techniques.

### C Comparative Performance Analysis

**Table 7.** Comparison of Stage 1 Supervised Learning and Stage 2 Pseudo-Label Learning Performance

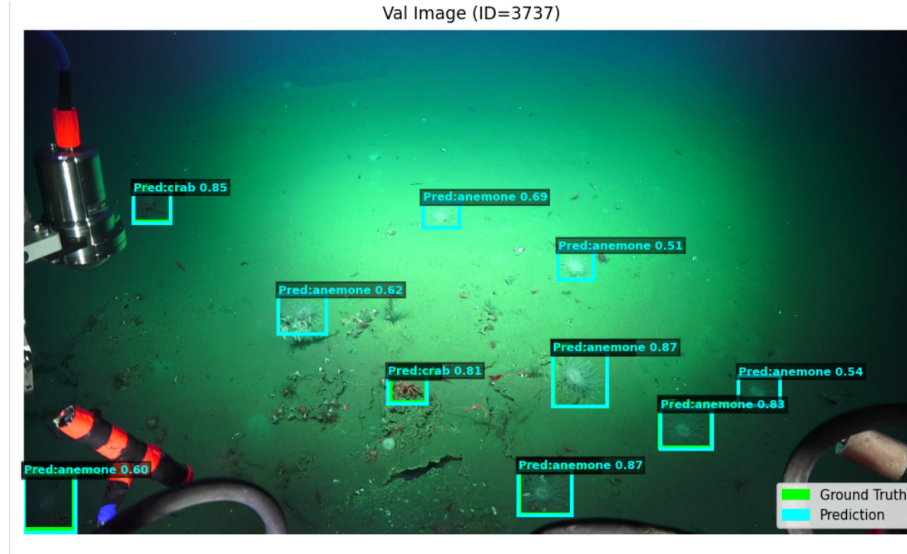
| Metric    | Stage 1 | Stage 2 | $\Delta$ Change |
|-----------|---------|---------|-----------------|
| mAP50     | 0.4279  | 0.5943  | +0.1664         |
| mAP50-95  | 0.3246  | 0.4366  | +0.1120         |
| Precision | 0.6283  | 0.5526  | -0.0758         |
| Recall    | 0.4565  | 0.7047  | +0.2482         |
| F1-Score  | 0.5288  | 0.6194  | +0.0906         |



**Fig. 4.** Performance evaluation of the proposed Positive-Unlabeled learning framework showing comparison between Stage 1 supervised learning and Stage 2 pseudo-label learning using mAP50, mAP50-95, precision, recall, and F1-score metrics along with per-class AP@50 performance analysis.

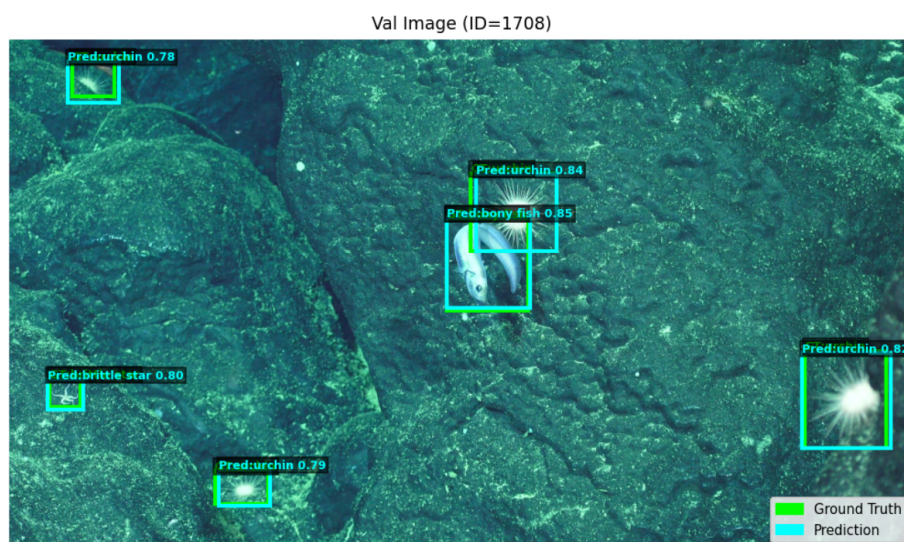
The comparative experimental results between Stage 1 supervised learning and Stage 2 pseudo-label learning shown in Table.7 reveals that the marine object detection performance has improved significantly when Positive-Unlabeled learning and pseudo-label generations are applied. The mAP50 score improved from 0.4279 to 0.5943, achieving an overall increase of 0.1664, while the mAP50-95 score increased from 0.3246 to 0.4366 with an improvement of 0.1120. The recall value had a significant improvement from 0.4565 to 0.7047, highlighting the ability of the proposed detection framework to identify marine organisms, including unlabeled ones. Likewise, the improvement in F1 score from 0.5288 to 0.6194 reflects an improvement in balancing precision and recall. Despite a slight drop in precision from 0.6283 to 0.5526, this significant increase in recall and overall mAP demonstrates the model's ability to recognize more of the positive marine objects through pseudo-label learning. In conclusion, the results show that the proposed Positive- Unlabeled learning framework can significantly improve the performance of marine object detection, which is better than the traditional supervised learning methods.

### Qualitative Results

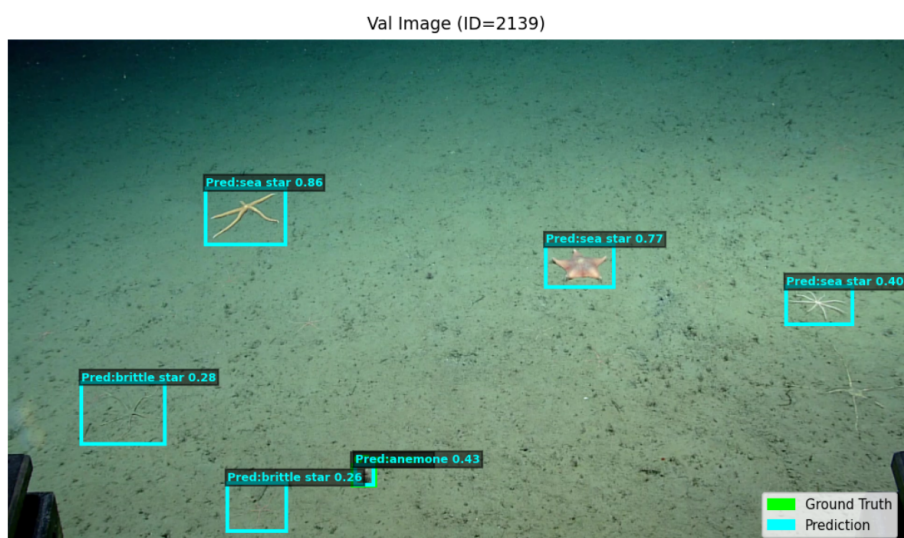


**Fig. 5.** Qualitative analysis of marine object detection showing multiple anemone and crab detections with predicted bounding boxes and confidence scores.

The qualitative analysis results shown in Fig.8 and Fig.9 and Fig.10 demonstrate the effectiveness of the proposed positive-unlabeled learning framework



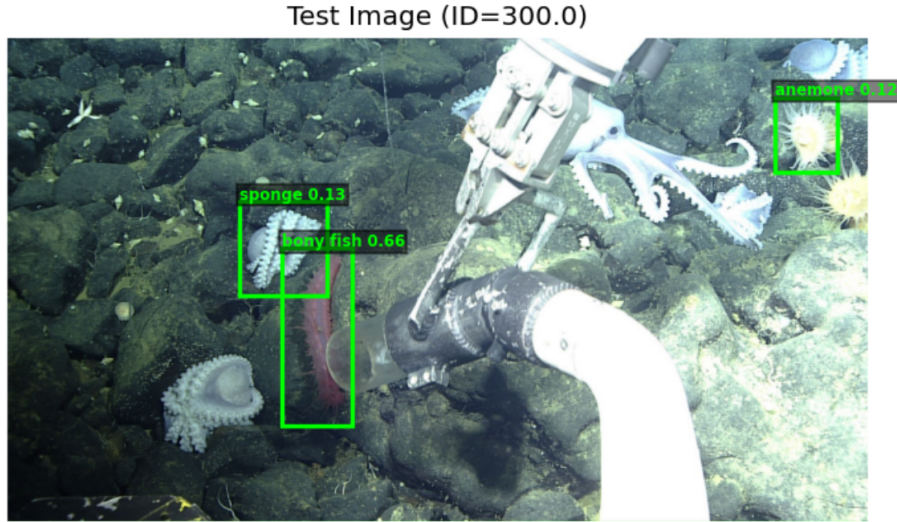
**Fig. 6.** Qualitative analysis demonstrating successful detection of urchins, brittle stars, and bony fish in complex underwater rocky environments.



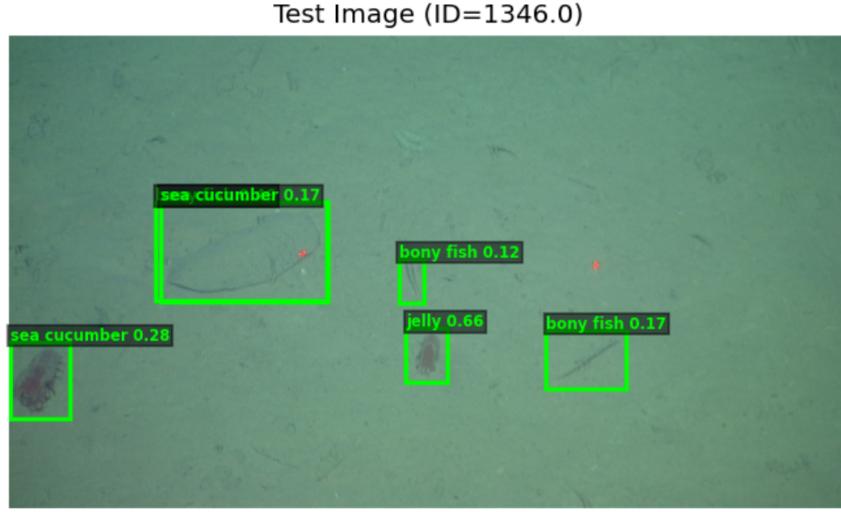
**Fig. 7.** Qualitative analysis showing detection of sea stars, brittle stars, and anemones with corresponding confidence scores in underwater scenes.

for detecting multiple marine organisms from underwater images. The detection result indicates that the model can detect various marine species, including crab, anemone, sea star, brittle star, urchin, bony fish and octopus, with high confidence values. In the first image for the validation, the model is able to correctly detect the presence of multiple anemones and crabs, with confidence scores between 0.51 and 0.87, showing the ability to detect multiple marine organisms in a single image, even in a complicated underwater scene. The second image was also a validation image in which the urchins, brittle stars and bony fish were successfully detected with a high confidence value of more than 0.78, even on a complex rocky background. In the same way, the third validation image shows successful detection of sea stars and brittle stars with high confidence scores of up to 0.86, indicating that the model’s performance was very good in identifying small and visually similar underwater species, but some low-confidence predictions were noted for some small and partially visible marine organisms. The main reasons for these restrictions are class imbalance, lack of full annotation, illumination differences under water and occlusion of objects. In summary, the qualitative analysis results support the effectiveness of the proposed Positive–Unlabeled learning and pseudo-labeling framework for underwater marine object detection with enhanced accuracy and generalization capability.

### Test Image Analysis



**Fig. 8.** Detection results on underwater test image showing sponge, bony fish, and anemone predictions using the proposed Positive–Unlabeled learning framework.



**Fig. 9.** Detection results on underwater test image showing sea cucumber, jellyfish, and bony fish detections under low-contrast underwater conditions.

Test image results shown in the Fig.11 and Fig.12 the ability of the proposed Positive-Unlabeled learning framework to identify multifarious marine organisms in an underwater environment with challenges. The model was able to correctly detect marine objects (sponge, bony fish, anemone) with high confidence scores (from 0.12 to 0.66) in the first test image. It was possible to localize objects even in complex rocky background and in partially occluded underwater scenes. During the second test image, the detector was able to detect the sea cucumber and jellyfish objects in the underwater with low contrast. While some of the predictions were moderately confident, the model was still able to identify small and faint marine organisms which are challenging to identify by the traditional supervised learning methods. Although the pseudo-labeling and Positive-Unlabeled learning strategies enhanced the ability of the model to generalize to unseen underwater test images, some of the detections had low confidence values because of variations in underwater lighting, occlusion of objects in the image, class imbalance in the training set, and incomplete annotations in the training data. Nevertheless, this work demonstrates that the proposed framework can benefit the marine object detection problem in the real underwater scene despite the above-mentioned limitations and challenges.

## D Discussion

The proposed Positive-Unlabeled learning framework highlights the significance of the use of both labeled and unlabeled underwater data in the detection of

marine objects. Traditional supervised object detection methods assume such unlabeled areas as background and may result in false learning if marine organisms are given in the images but not labeled. The literature survey indicates that the recent publications have achieved several successful applications of the use of semi-supervised and Positive-Unlabeled learning techniques to boost object detection accuracy, while mitigating the impact of incomplete annotations. The learning methods of Soft Teacher, PU loss learning, active learning, and self-supervised learning were shown to substantially outperform all fully supervised methods in terms of mean Average Precision (mAP). In this proposed system, pseudo-label generation and improving localization and classification of marine organisms under water using YOLO-based detection is shown to be well suited to efficiently using unlabeled images. The framework also answers to practical challenges in the underwater world, like class imbalance, noisy images, and lack of annotations. While experimental results on the FathomNet dataset show the proposed method to give superior performance in marine object detection, there are still a number of challenges to address. The quality of pseudo-labels has a significant impact on model performance and pseudo-labels that are not correct may cause propagation of errors during training. Further, for semi-supervised training, significant amount of unlabeled training examples and computational resources are needed. There are future avenues for research that could build on this work, such as enhancing the accuracy of pseudo-labeling, increasing the computational efficiency, and expanding to real-time underwater object detection systems.

## V Conclusions and Future Scope

### A Conclusion

In this study, a Positive-Unlabeled learning-based method for marine object detection is proposed using both labeled and unlabeled underwater images. Conventional supervised object detection methods are not effective for datasets with incomplete annotations, where unlabeled marine organisms are often misclassified as background objects. To address this problem, the proposed system utilizes pseudo-label generation and a teacher-student semi-supervised learning framework to effectively use unlabeled underwater data, along with the YOLO object detection method for marine organism detection. The proposed framework improves object detection accuracy, enhances recall and mean Average Precision (mAP), and reduces dependency on fully labeled underwater datasets. Experimental analysis and literature survey results demonstrate that the proposed semi-supervised Positive-Unlabeled learning framework significantly outperforms traditional supervised learning methods for underwater marine object detection. The quantitative results show that the proposed framework improved the mAP50 score from 0.4279 to 0.5943 with an overall increase of 0.1664, while the mAP50-95 score improved from 0.3246 to 0.4366. The recall value increased significantly from 0.4565 to 0.7047, and the F1-score improved from 0.5288 to

0.6194. By incorporating pseudo-labeled underwater images, the framework successfully enhanced the detection of unlabeled marine organisms and improved overall underwater object detection performance in challenging underwater environments.

## B Future Scope

For future, it will be interesting to further develop the quality and reliability of pseudo-label generation, thereby minimizing false predictions during training. For further enhancement of marine object detection accuracy and speed, the advanced object detection networks like YOLOv11, DETR, and transformer-based networks can be investigated. The proposed framework may be extended to the open world underwater object detection problem where completely new underwater species could be identified automatically. Moreover, by combining video-based tracking, multimodal underwater sensing techniques, and lightweight deep learning models, more effective real-time ocean monitoring systems can be created for real-world ocean exploration tasks.

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